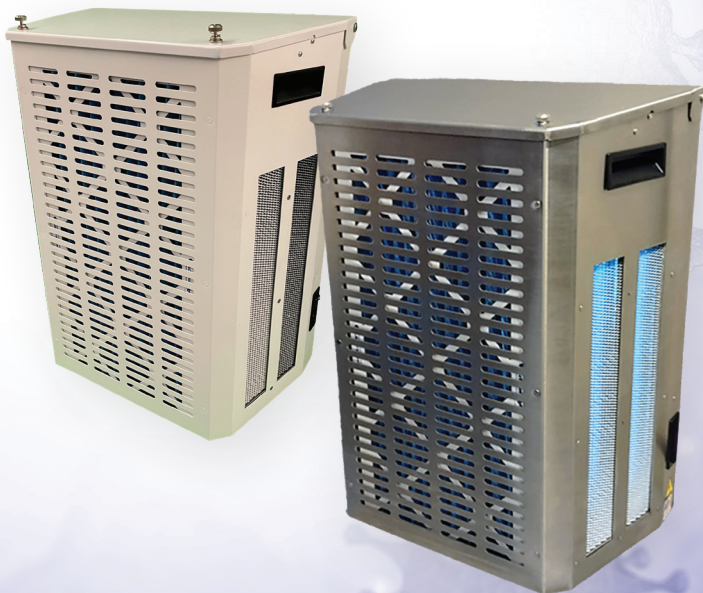


MAK

MICROBIAL AREA KLEANER

MAK-414™ Owner's Manual



MAK-414™

MICROBIAL AREA KLEANER™



Warnings and Precautions

WARNING: ALWAYS DISCONNECT POWER TO THE UNITS BEFORE PERFORMING ANY MAINTENANCE.

Follow all installation instructions. Failure to install unit properly may lead to injury or unit not working properly.

Use this unit in a manner intended by the manufacturer. If you have questions, contact the manufacturer.

RISK OF ELECTRIC SHOCK: These servicing instructions are for use by qualified personnel only. To reduce the risk of electric shock, do not perform any servicing other than that contained in the operating instruction unless you are qualified to do so.

THIS PRODUCT SHALL NOT BE INSTALLED BEHIND A SUSPENDED FLOOR/CEILING OR A STRUCTURAL WALL, CEILING OR FLOOR.

Before performing any service or cleaning of the unit, unplug it from its power source.

Replace filters regularly to avoid damaging your unit.

WARNING: This appliance is equipped with a three prong (grounding) plug. Product will malfunction and/or cause shock if not grounded. This product should be plugged directly into a properly grounded three prong receptacle.

WARNING: This device produces potentially harmful UV light. Disconnect from power supply before servicing. Failure to follow warning may cause damage to skin and eyes.

THE HEALTH ASPECTS ASSOCIATED WITH THE USE OF THIS PRODUCT AND ITS ABILITY TO AID IN DISINFECTION OF ENVIRONMENTAL AIR HAVE NOT BEEN INVESTIGATED BY UL.

Our products are not medical devices and are not intended to diagnose, treat, or cure any disease. Extreme Microbial Technologies, LLC declines all responsibility for all damages arising from improper use of any unit or in the case of tampering with the unit.

CONGRATULATIONS on your purchase of a MAK-414™ Microbial Area Kleaner™. This unit has been designed and engineered for easy installation onto a wall and used as an air and surface purification system. Each unit creates Energized Hydrogen Peroxide molecules which circulate throughout the indoor space.

Energized Hydrogen Peroxide molecules are safe for people, pets and plants and seek out and neutralize a host of microbial contaminants such as molds, bacteria, and viruses.

The MAK-414™ will provide years of trouble-free service when properly operated and maintained. This guide has been put together to help you through the steps required to install and operate your unit. The air filters will need to be replaced periodically and the cells will need to be replaced at least every 24 months.

Product Specifications

Product	MAK-414
Weight (lbs)	41
Dimensions (Inches)	17 x 9.5 x 21
Power (Type)	AC
Power (Watts)	50
Power (Volts)	120
Current (Amps)	4.2
Warranty	1 year
Filters	1 Main filter
Filter Life (Approx.)	4-8 weeks
Air Flow (CFM Max)	500-800*
Coverage (CF)	
Typical Coverage	48,000-72,000
High Bioload Coverage	12,000-24,000

*Coverage Range is affected by microbial load, physical layout, temperature, humidity, air exchange and air circulation patterns.

Installation Tools and Parts Needed for Wall Mounting

Tools Required: Drill, Bubble Level, $\frac{3}{16}$ " Drill bit, $\frac{1}{4}$ " Drill bit, Phillips Screwdriver

Hardware Supplied: 4 - #10-12 (M5) Pan Head Screw and 4 - Plastic Anchors

NOTE: 4 screws are supplied if you want to use both mounting brackets.

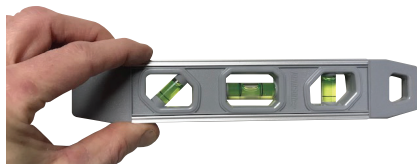
Installation with Mounting Bracket (Two Screws)

Step 1. Identify the location you wish to mount your MAK™ unit. Mark the area with an awl or pencil. (Optional - Use an electronic stud finder to determine location of studs in the wall.)

Step 2. Mount the wall-hanging bracket provided with 3 screws and anchors. Green circles show the holes that you will use for mounting.



Step 3. Use a bubble level to mark a level line on the wall at the height you wish to mount the unit. Measure and mark the locations for screws to go into wall for mounting. You will be marking three (3) holes. Use the mounting bracket as a template on the wall, using the bubble level, to mark the location for drilling the holes.



Step 4. Pre-drill a $\frac{3}{16}$ " (4.5mm) hole to a depth of 1 $\frac{1}{4}$ " (32 mm) into the wood stud at each marked location. If there is no stud enlarge the hole using a $\frac{1}{4}$ " drill bit and insert a plastic anchor for #6 (M4) screw into one or all holes on the wall.



Step 5. Use the three (3) screws provided to mount the wall-hanging bracket securely onto the wall. Leave the screw head out approximately $\frac{1}{4}$ " (6 mm) from the wall to allow for the bracket and keyhole to set onto the screw. Make sure it is level and firmly tightened flush against the wall.

Step 6. Hang the unit on the wall by placing the top bracket over the two (2) hooks on the wall-hanging bracket.

Step 7. Plug the power cord into the unit and then into the power outlet on the wall.

Placement Guidelines

Place the unit off the ground and ensure there are no obstructions to the air flow.

Place the unit so that the Energized Hydrogen Peroxide molecules are distributed into the existing room air circulation pattern to ensure complete disbursement throughout the room.

Do not place the unit near any return air ducts. Ideally place the unit on an opposite wall.

Place the unit so that water does not get onto or into the unit.

Cleaning and Maintenance

Unplug your unit from the electrical outlet before any cleaning or maintenance is done.

Each MAK-414™ unit should have the filter cleaned and/or replaced every 1-2 months depending on the quality of the air in the environment. Each unit uses a main filter.

To change filter:

1. Unplug unit.
2. Use fingers to hook into the vent holes at the top of the filter cover and pull out to release. The filter cover has a magnetic catch at the top and two placement pins at the bottom.
3. To remove old filter pad, slide the filter up into the gap in the top part of the filter cover so that the bottom part of the filter can be lifted away from the cover. Pull the filter out of the top of the cover.
4. Reverse the process to install new filters. Pre-filter goes in the front.
5. Reinsert placement pins at the bottom of the cover back into the unit and push the top of the filter cover back into the unit.

If the filter cavity has dust or debris inside, we recommend using an Air Duster (or similar) to gently blow the debris out of the unit. If any debris remains, wipe it down with a damp cloth and let dry completely.

NOTE: For optimal performance we recommend changing the cell every 24 months even if the UV lamp is glowing and appears to be operating normally.

Changing the MAK-414™ Cells



Warning: UV light in operation. Disconnect from power supply before servicing. Failure to follow warnings could result in serious temporary eye or skin irritation.



Warning: Risk of electrical shock. Always disconnect power to the unit before performing any maintenance to the unit.

1. Turn off unit, unplug and remove from wall before starting to change cells.
2. Loosen the bolts and open the door at the top of the unit.
3. Use a flathead screwdriver (as shown below) to remove the metal plate covering the cells.
4. Grasp top of the bulb with one hand and lamp connector with the other hand and gently disconnect the lamp from the connector.
5. Lift each cell one at a time out of retaining bracket.



6. Place cell back into the lamp bracket making sure the **angled**

side of honeycomb is facing out and downward.

7. Attach the lamp connector to the new bulb.
8. Replace the metal plate back over the cells.
9. Place cover back in place and re-tighten the bolts.
10. Re-hang the unit, plug back in, and turn back on after cells have been changed.

Cell Disposal

Please do not throw your cell into the garbage. The special lamp used within the MAK™ cell contains a very small amount of mercury. Be sure to dispose of lamps properly or return your used cell to Extreme Microbial Technologies for proper disposal.

Send your used cell postage paid to:

Extreme Microbial Technologies, LLC
2800 East River Road, Suite A
Dayton, OH 45439

LIMITED WARRANTY INFORMATION

Your MAK-414™ unit is warranted to be free of all defects in material and workmanship in normal use for a period of 1 year.

The warranty is granted only to the original purchaser. The warranty is subject to the following provisions:

Any damages or malfunctions caused by negligence, abuse, or use not in accordance with this Product Owner's Manual are not covered by this warranty. Likewise, any defects or damages caused by unauthorized service or the use of other than genuine Extreme Microbial Technologies, LLC parts are not covered.

Extreme Microbial Technologies, LLC will, at its option, repair or replace a defective product or part(s) for the product that is/are covered by this warranty. As a matter of warranty policy, Extreme Microbial Technologies, LLC will not refund the customer's purchase price.

OBTAINING WARRANTY SERVICE

To obtain warranty service you must return the defective product along with proof of purchase to Extreme Microbial Technologies, LLC. All shipping costs submitted under this warranty shall be paid by the purchaser. Unless the warranty is expressly renewed or extended in writing by Extreme Microbial Technologies, LLC any repaired or replaced part of the unit shall be warranted to the original purchaser only for the length of the unexpired portion of the original warranty.

Before any product is returned for service, the customer should contact Extreme Microbial Technologies to obtain a Return Merchandise Authorization (RMA) Number and a return shipping address. The RMA Number should be clearly written on the box before shipping. All parts/components, manuals, and original packaging should be included in the return if available.

FURTHER LIMITATIONS AND EXCLUSIONS ARE AS FOLLOWS

Any warranty that can be implied in connection with your purchase or use of this product, including any warranty of merchantability or any warranty of Fitness for a Particular Purpose is limited to the duration of this warranty. Some states do not allow limitations on how long an implied warranty lasts, so the above limitations may not apply to you.

Your relief for the breach of this warranty is limited to the relief expressly provided above. In no event shall Extreme Microbial Technologies, LLC be liable for any consequential or incidental damages you may incur in connection with your purchase or use of this product. Some states do not allow the exclusion or limitation of incidental or consequential damages, so the above limitation or exclusion may not apply to you.

This warranty gives you specific legal rights, and you may also have other rights which vary from state to state.



2800 E River Road, Suite A
Moraine, OH 45439

extrememicrobial.com